CNC Procedure 2016_016

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Touch Panel Calibration

[**Applicable for Fanuc Series 16/18/21/30/31/32/35**]

About Touch Panel

A touch pad can be used on the LCD display, as follows:

1. The soft keys (Ver - F0 to F9, Hor - F0 to F9, FR, and FL) in the lower and side section on the color LCD/MDI panel are changed to those for the touch pad.
2. Touch pad operation substitutes for cursor control on the color LCD.
3. A touch pad can be used to create C executor--based applications such as touch pad--based software versions of a machine operator's panel and a calculator.

Hardware characteristics

The hardware characteristics of the touch pad are briefly described below for reference purposes. The user need not be aware of them when using the touch pad. When a point on the touch pad is pressed, the touch pad controller sends the coordinates of the point to the CNC via an RS--232C interface.

A. Coordinate system:

1. Initial coordinate system
   The upper left corner of the input area of the touch pad is set as the origin (X = 0, Y = 0) of the coordinate system. The maximum coordinates of the lower right corner is (639, 479).
2. Coordinate system after compensation
   The coordinate system can be subjected to compensation as described later, so the initial origin and maximum coordinates can be maintained.

B. Reception Data sampling interval:

1. The coordinates of a point on the touch pad are output via RS--232C interface.
   The information is received at sampling intervals of 50ms.

C. Positional Precision:

1. A positional precision of ±2.5mm can be maintained by performing 9--point compensation as described after.

Software key control

When a rectangular soft key button on the screen is pressed with a finger, the soft key button display is indented. This type of a soft key behaves in the same manner as conventional soft keys. In addition, the touch pad soft keys can be customized using the C executor.

Cursor control

When the cursor is displayed and can be moved (for data setting) on the LCD panel with a touch pad, pressing the desired point on the screen causes the cursor to move to that point, in the same way as with the below keys:

Why Compensation is required?

1) Specifies an effective input area.
2) Specifies maximum coordinates.
3) Corrects linearity.
4) Makes the panel input position match the LCD display position.
5) Execution time

When Compensation must be executed?

1) A new touch pad is installed.
2) An existing touch pad is replaced.
3) Memory (SRAM) is cleared to all 0s.

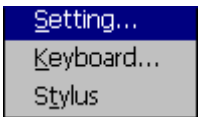
What to take care during compensation?

1) If an alarm condition occurs during compensation, the alarm screen appears when the compensation screen is exited.
2) Compensation can be made regardless of what the current mode is.
3) Once the system gets started, compensation should be made promptly before actual operations begin.

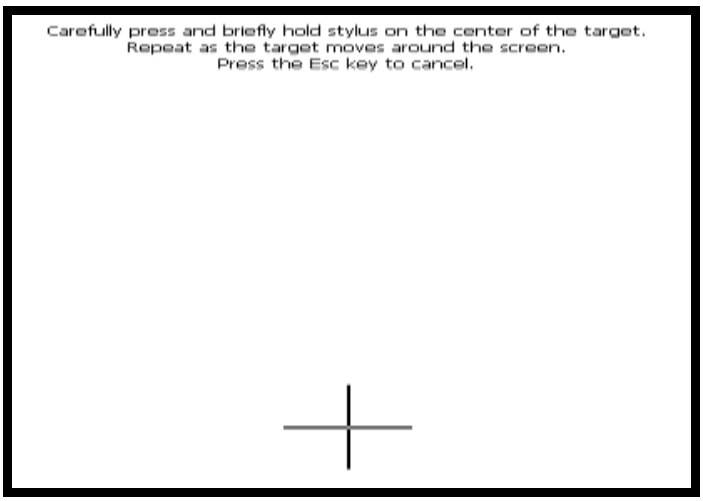
How to do compensation?

Method – 1 (*Applicable for TP with CNC display unit with PC functions FS160/180/210/300/310/320 i A/B controllers):*

1. Press the **[ALT-O]** key to display the following function menu.



2. Select **[Stylus]** to display the calibration screen.



3. When the **crosshair cursor** appears, press and hold the center of the cursor until it disappears (about three seconds).

4. When the cursor position moves, perform step 3 again. Repeat this process a total of five times.

5. Press the **RETURN** key to close the calibration screen. To cancel the settings of the touch panel, Press the **ESC** key.

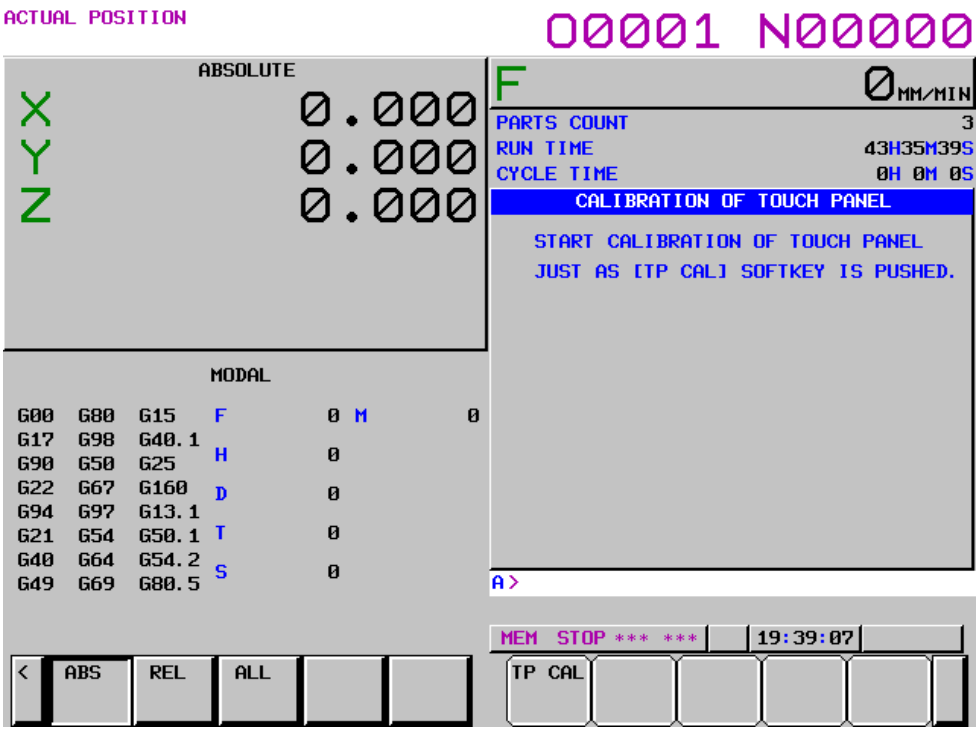
6. Please check for touch point responding and respective screen or function is executed

End of Procedure.

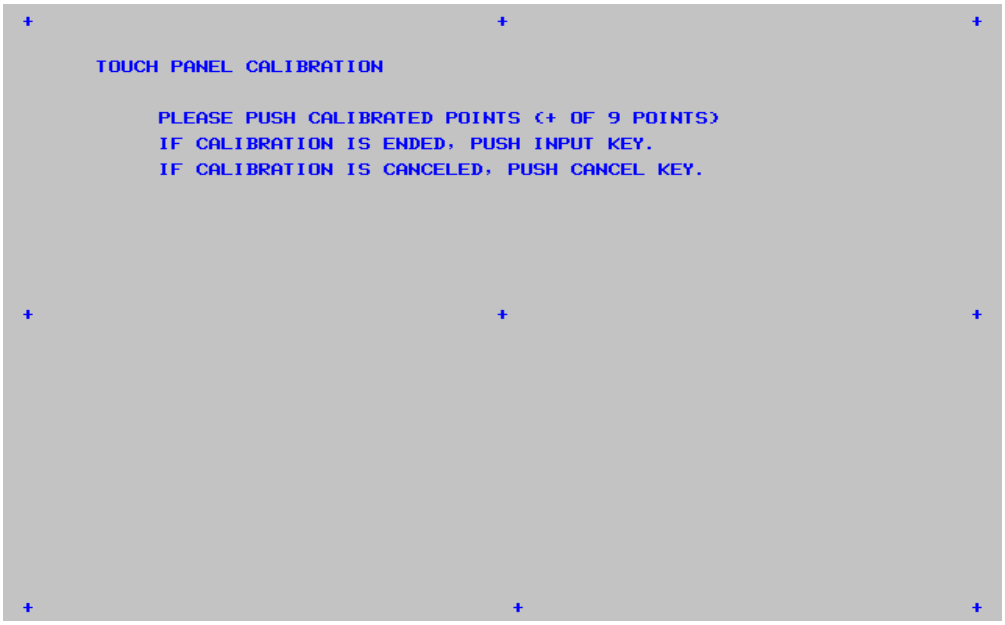
Method – 2: (*Applicable for FS16/18/21/30/31/32i A/B controllers*)

Calibration procedure:

- Enable the touch panel calibration screen (Set bit 5 (DCL) of parameter No. 3113 to 1.)
- Press function key SYSTEM
- Press the continuous menu key ⇨ several times.
- The [TOUCH PANEL] soft key is displayed.



- Press the [TOUCH PANEL] soft key then the [(OPRT)] soft key.
- The [TP CAL] soft key is displayed.



- Press the calibration points (9 points) with a dedicated pen. When a point is pressed correctly, the "+" mark changes to the "O" mark. If a "+" mark is not pressed accurately, the message "CALIBRATION POINT DOES NOT MATCH. PLEASE PUSH AGAIN." is displayed.the [TP CAL] soft key displays the touch panel calibration screen for all screens.

- Press After entering the calibration points (9 points), press the <INPUT> key to complete calibration. To cancel calibration or retry, press the <CAN> key. The screen display returns to the previous screen. If the <INPUT> key is pressed before entering the calibration points (9 points), calibration operation is cancelled.

- When calibration is terminated normally, the message "CALIBRATION WAS ENDED." is displayed.

- Upon completion of calibration, disable the touch panel calibration screen to protect against operation errors. (Set bit 5 (DCL) of parameter No. 3113 to 0.)

NOTE

I. If the screen display is switched from the touch panel calibration screen to another screen such as the alarm screen by the automatic screen switching function, calibration is automatically cancelled.

II. Calibration operation can be performed in any mode.

III. After system startup, perform calibration operation swiftly before starting operation.

Parameter

	#7	#6	#5	#4	#3	#2	#1	#0
3113			DCL					

[Input type] Parameter Input

[Data type]Bit

 # 5 DCL The touch panel calibration screen is:

0:Disabled.

1:Enabled.

Usually, set this parameter to 0. Touch panel calibration becomes necessary only when the touch panel is replaced or all memory data is cleared. Set this parameter to 1 only when performing touch panel calibration. Upon completion of touch panel calibration, reset this parameter to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
3119						DDS		

[Input type] Parameter input [Data type] Bit

 # 2 DDS The touch panel is:

0:Enabled.

1:Disabled.

Set this parameter to 1 when disabling the touch panel temporarily as in the case of startup

Alarm and message

Number	Message	Description
SR530 3	TOUCH PANEL ERROR	The touch panel is not connected correctly, or the touch panel cannot be initialized when the power is turned on. Correct the cause then turn on the power again.

CAUTION

1. When all memory (SRAM) data is cleared, the soft keys on the touch panel are not made usable yet. In this case, the MDI keys (such as the cursor keys and page keys) need to be used for setting.
2. When two or more points are pressed at a time, the pressing of a weighted point determined by considering how all points are pressed is assumed. Be sure to press one point only at a time.

How to protect the touch panel?

Please use the touch panel protection sheet and touch pen
Example - For 15” Screen

Touch panel maintenance parts view with specification

Protection sheet for touch panel 15” LCD type A08B-0082-K020
Pen for touch panel - A02B-0236-K111 (Common for all touch panels)

LCD with a touch panel

The touch panel is operated by directly touching the LCD screen. For this operation, be sure to use a FANUC-supplied pen (A02B-0236-K111) dedicated to the touch panel. If a sharp-pointed pen is used, for example, to touch the LCD screen, the LCD surface may be flawed or damaged. Moreover, do not touch the LCD screen directly with a finger. Otherwise, the operability of the LCD may deteriorate, and the LCD screen may get dirty.

Protection sheet for the touch panel

A protection sheet is attached the face of an LCD with a touch panel to protect the thin film of the touch panel and LCD. If the protection sheet is damaged, it can be replaced. (The protection sheet is a consumable part.)

Replacing the protection sheet - Materials used

(1) Protection sheet A08B-0082-K020 For 15.0” LCD with touch panel
(2) Neutral detergent (detergent that can clean oily dirt off = detergent for kitchen can be used)
(3) Soft cloth (such as towel)

• Replacement procedure

- Turn off the power to the machine.
- Peel off the old protection sheet from the surface of the touch panel.
- Wipe off adhesive residue if any on the screen surface with alcohol.
- Use the detergent to remove oil or dirt stuck to the surface of the touch panel.
- With a soft, damp cloth, wipe off detergent completely. If the touch panel surface becomes cloudy, oil is still left on the surface. Remove oil completely. If oil or detergent is left on the surface of the touch panel, the protection sheet cannot adhere to the panel completely and will sometimes peel off easily.
- With a dry soft cloth, wipe off moisture completely.

*The periodic maintenance screen shows the current statuses of those consumables that require periodic replacement (backup battery, LCD backlight, **touch pad**, etc.). An item whose service life has expired is indicated by the machine run time or they like.*